

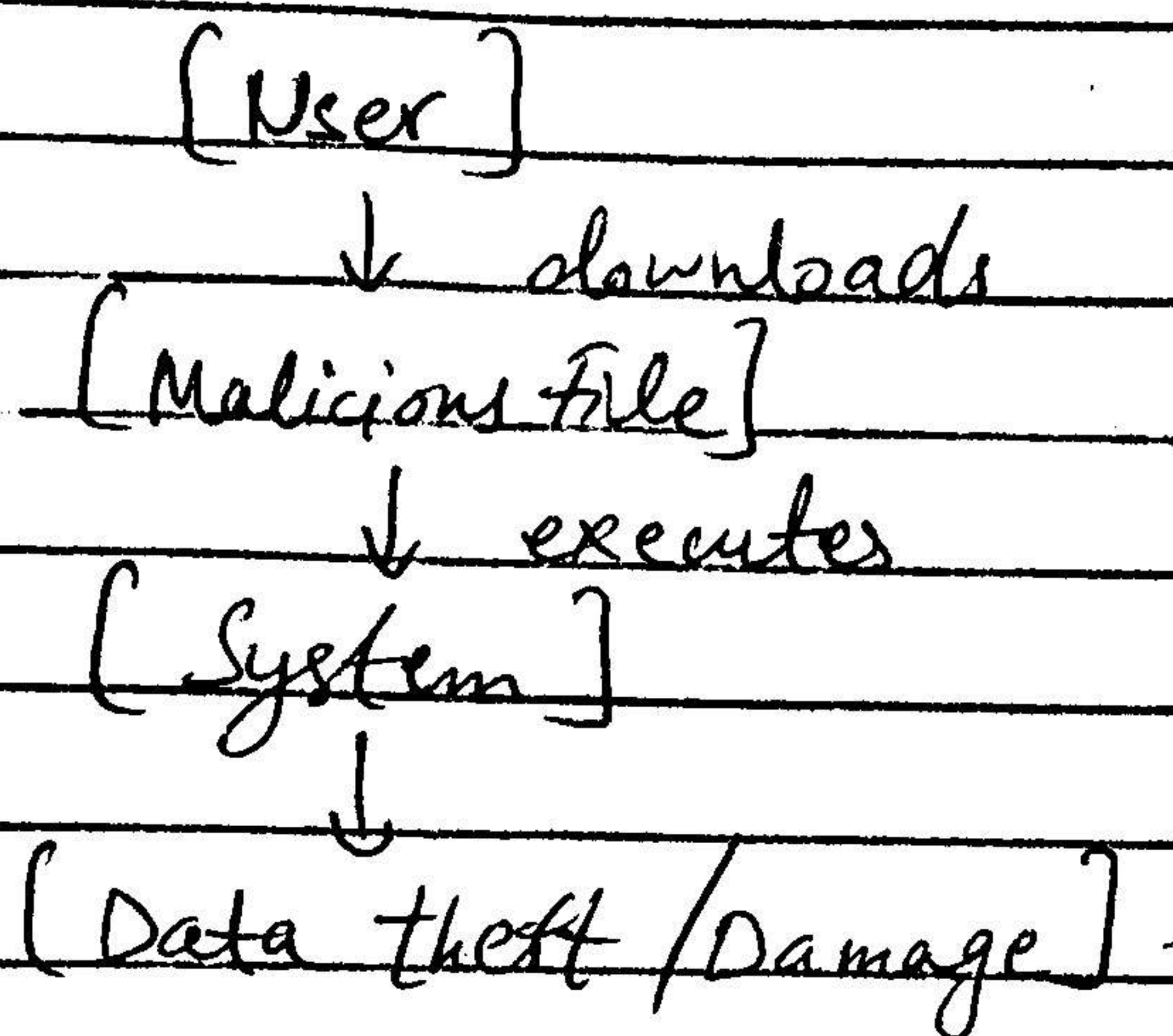
1) Malware Attacks:-

Working: It enters system through malicious files or downloads and execute harmful actions like stealing or damaging data.

Security: Antivirus, keep systems updated, avoid untrusted files.

Environment: User devices, enterprise systems, and network with weak endpoint security.

Diagrams



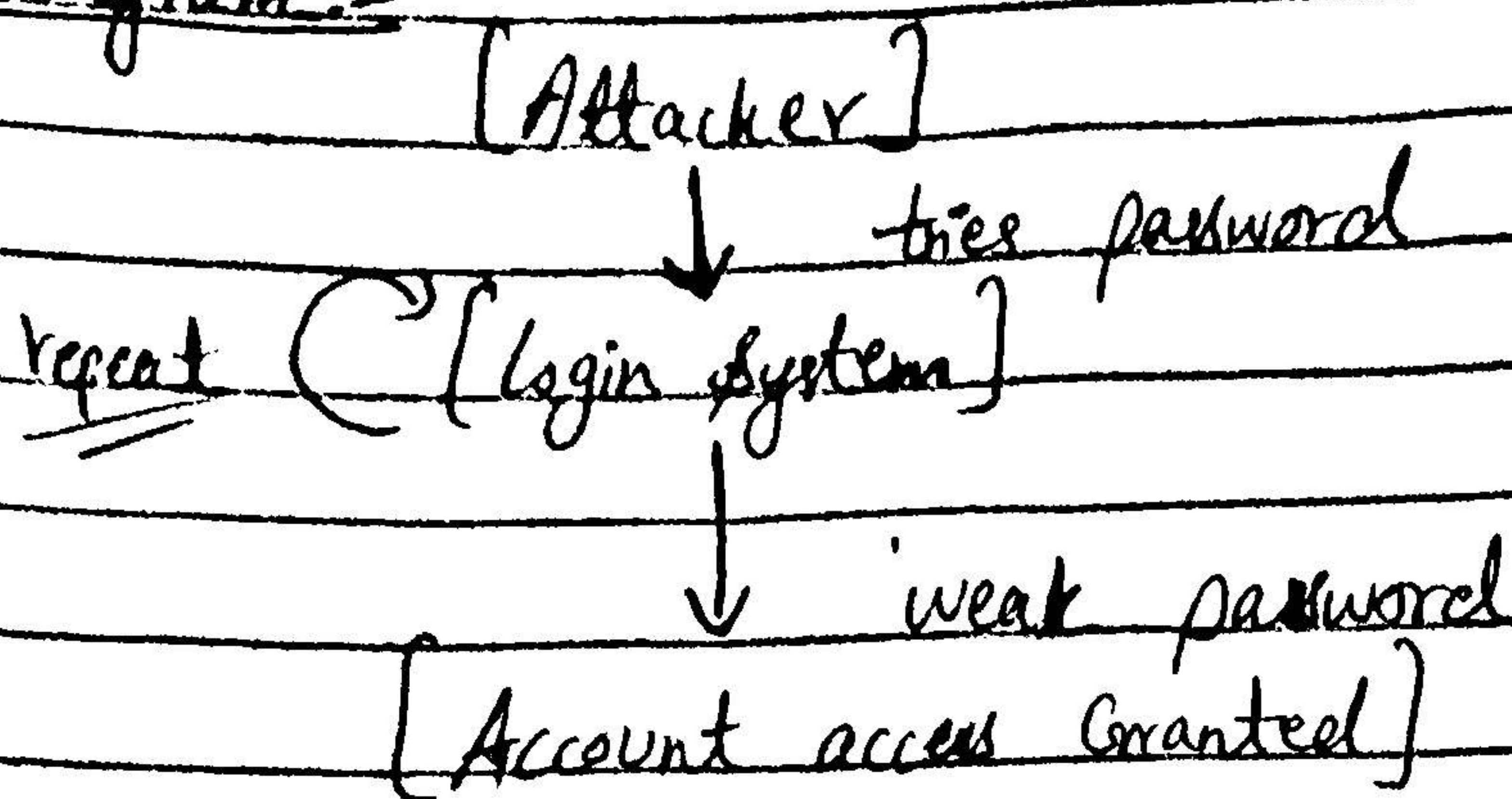
2) Password Attack:-

Working: The attackers repeatedly tries passwords using brute-force or dictionary attacks until access.

Security: Strong passwords, enable MFA, and account lockout policies.

Environment: Login systems, websites, and weak authentication platforms.

Diagram:-



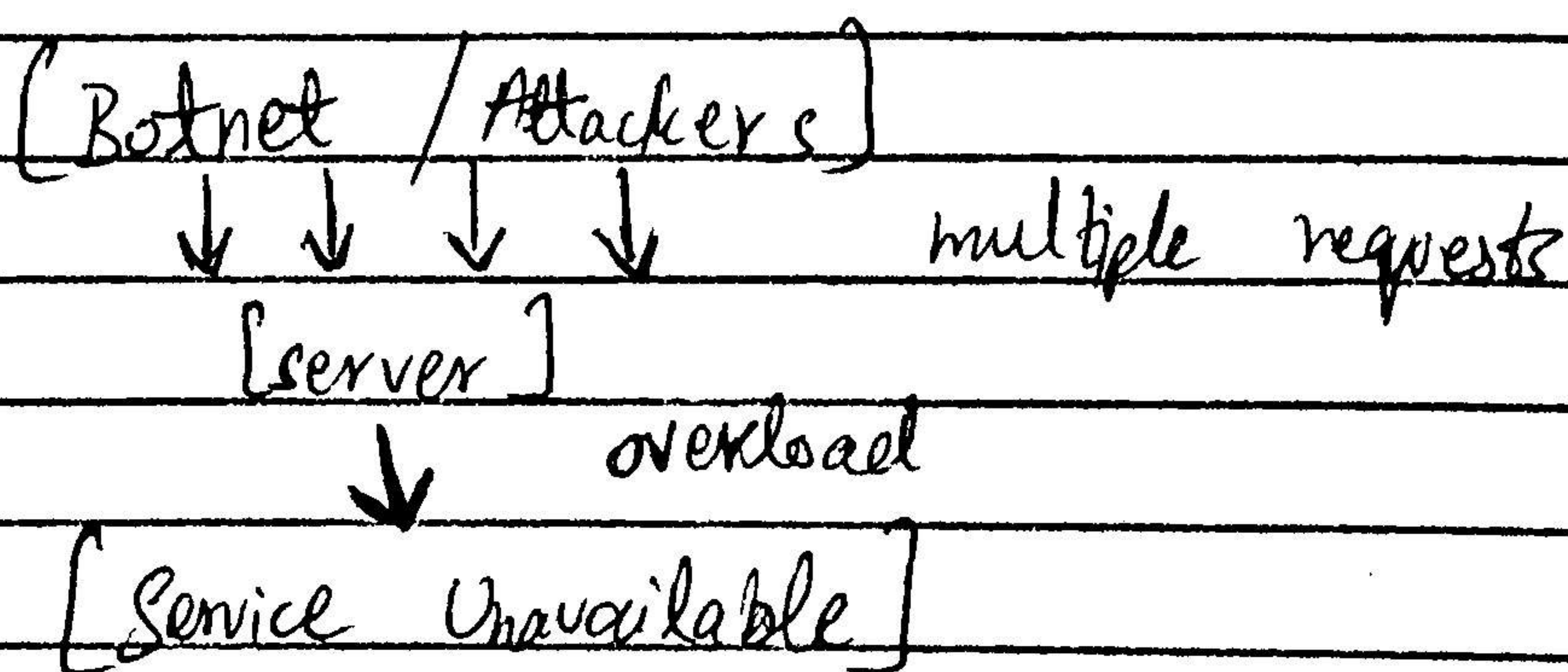
3) DOS/DDOS:-

Working: The attacker flood the server with excessive requests, making it unavailable to real users.

Security: Apply rate-limiting, firewalls, and traffic filtering / CDN protection.

Environment: Public websites, APIs, and cloud services.

Diagram:-



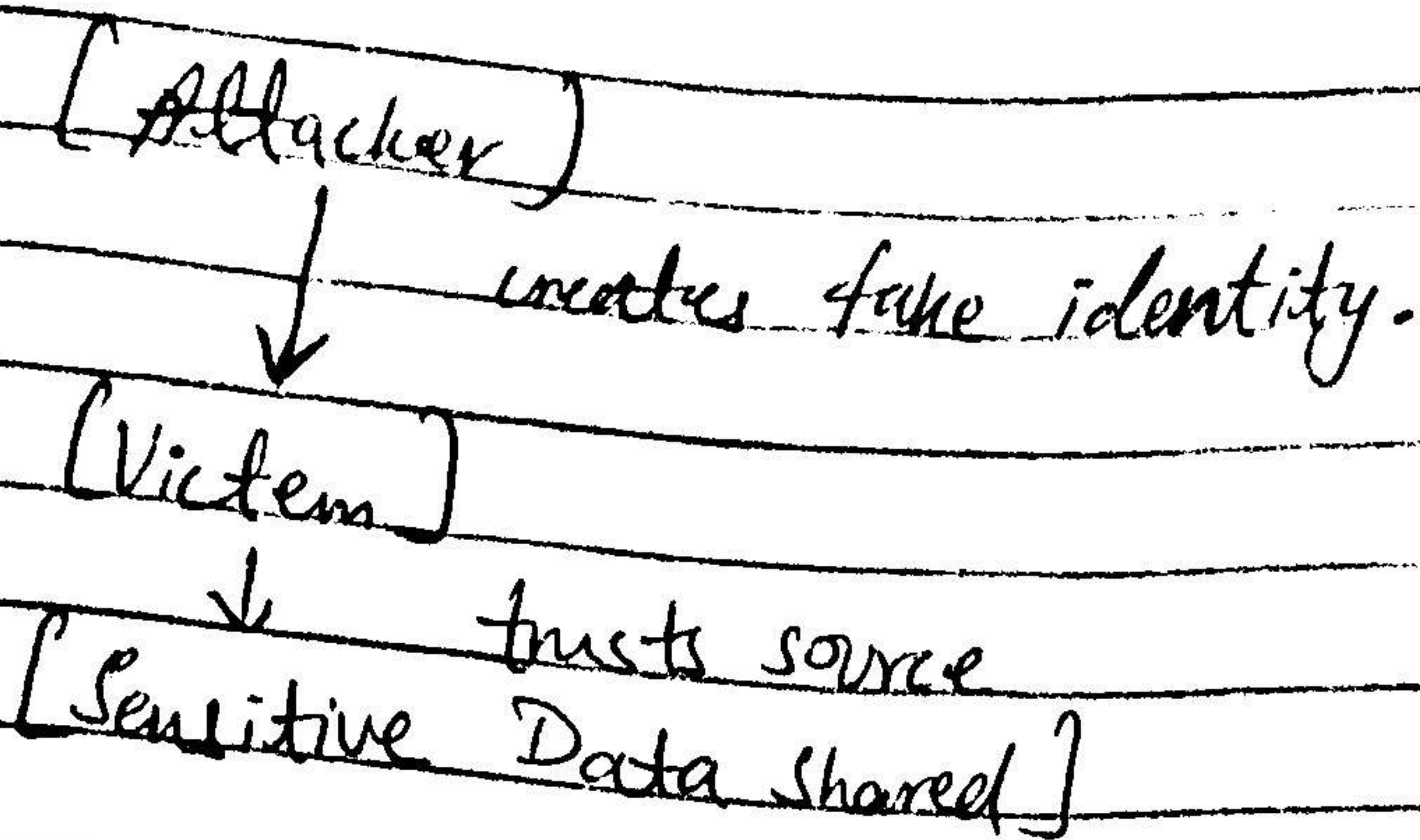
4) Spoofing:-

Working: Attackers pretend to be a trusted source to trick users into sharing data.

Security: Authentication protocols like SPF/DKIM and verifies identities

Environment: Email systems and network communications.

Diagram:-

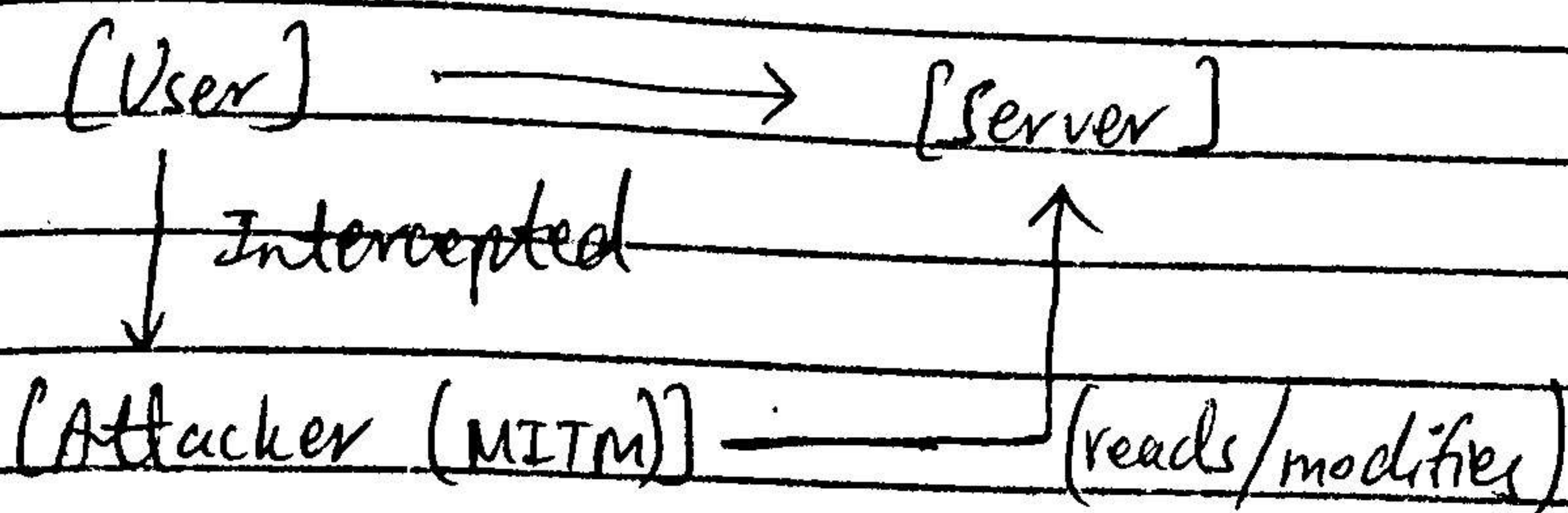


3) MITM:-

The attacker intercepts communication between two parties to read or modify data.

Security: Use HTTPS, encryption and VPNs

Environment: Public Wifi and insecure networks

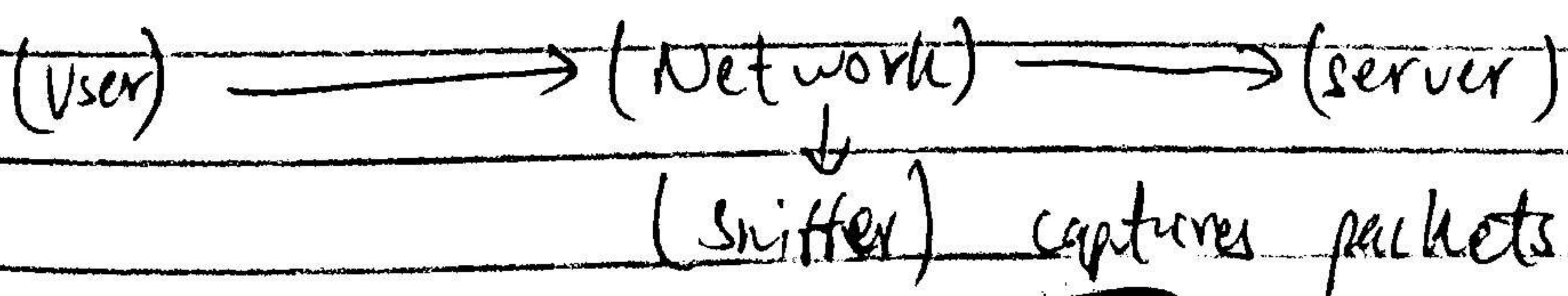


6) Sniffing

The attacker captures network packages to extract sensitive information.

Security: Use encryption (HTTPS / VPN) and secure network data.

Environment: Open networks and shared LANs.



7) Social Engineering:-

Psychologically manipulation of users to reveal confidential info.

Security: Provide user training and enforce verification procedures.

Environment: Organizations with untrained or unaware users.

[Attacker]

↓ manipulates

[User] reveals info [Confidentiality Compromised]

8) Phishing:-

Redirection of users to fake websites to steal login credentials.

Security: DNS security (DNSSEC) and HTTPS verification

Environment: Browsers and DNS systems

[User]

↓ enters URL

[DNS Redirect]

↓

[Fake Website]

↓

[Credentials stolen]

a) Spam Bombing

Sending a large number of emails to overload or disrupt systems.

Security: Spam filters and rate limiting.

Environment: Email servers and inbox systems.

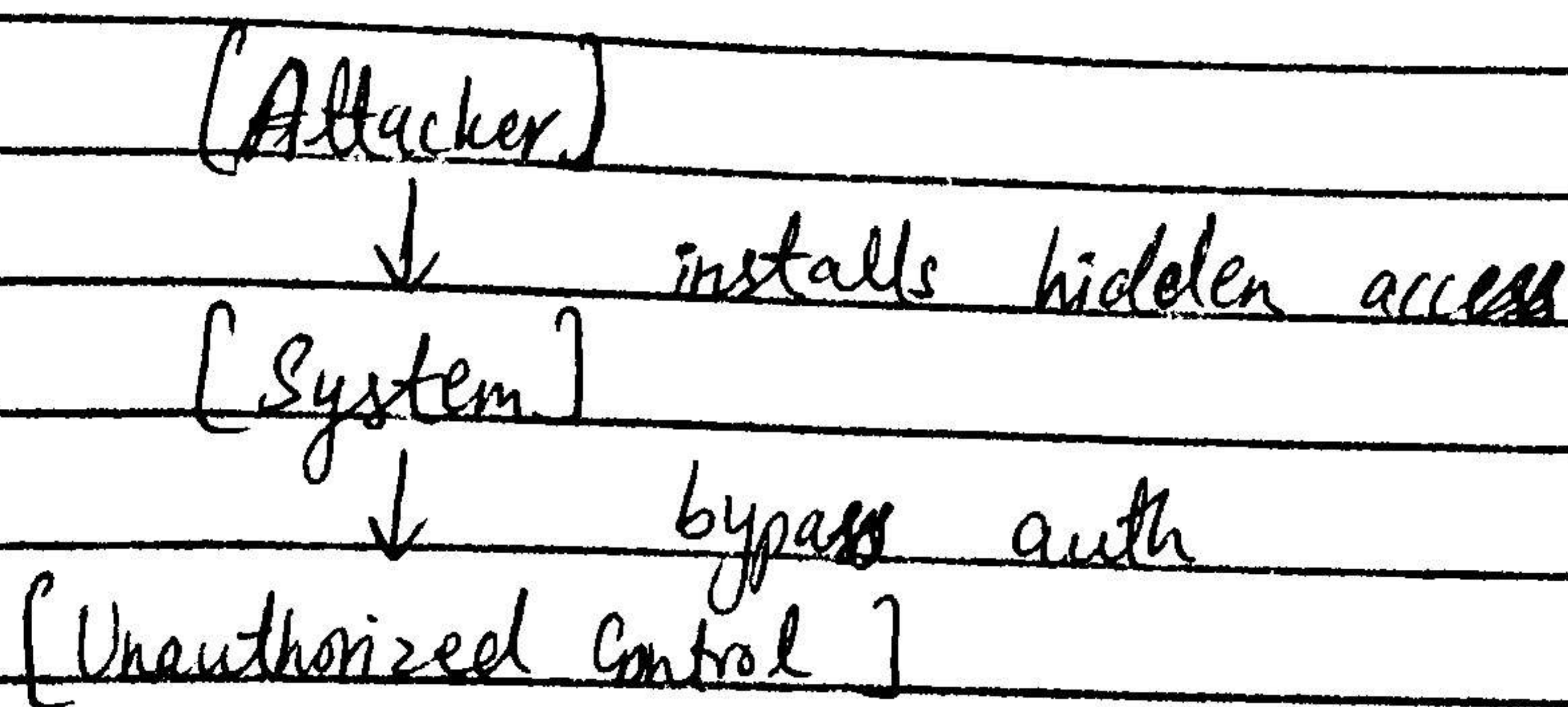
[Attacker] → Many Email → Overload.

b) Backdoor

The attacker installs hidden access to bypass normal authentication.

Security: Monitor systems, remove unknown accounts, and update software.

Environment: Compromised systems and servers.

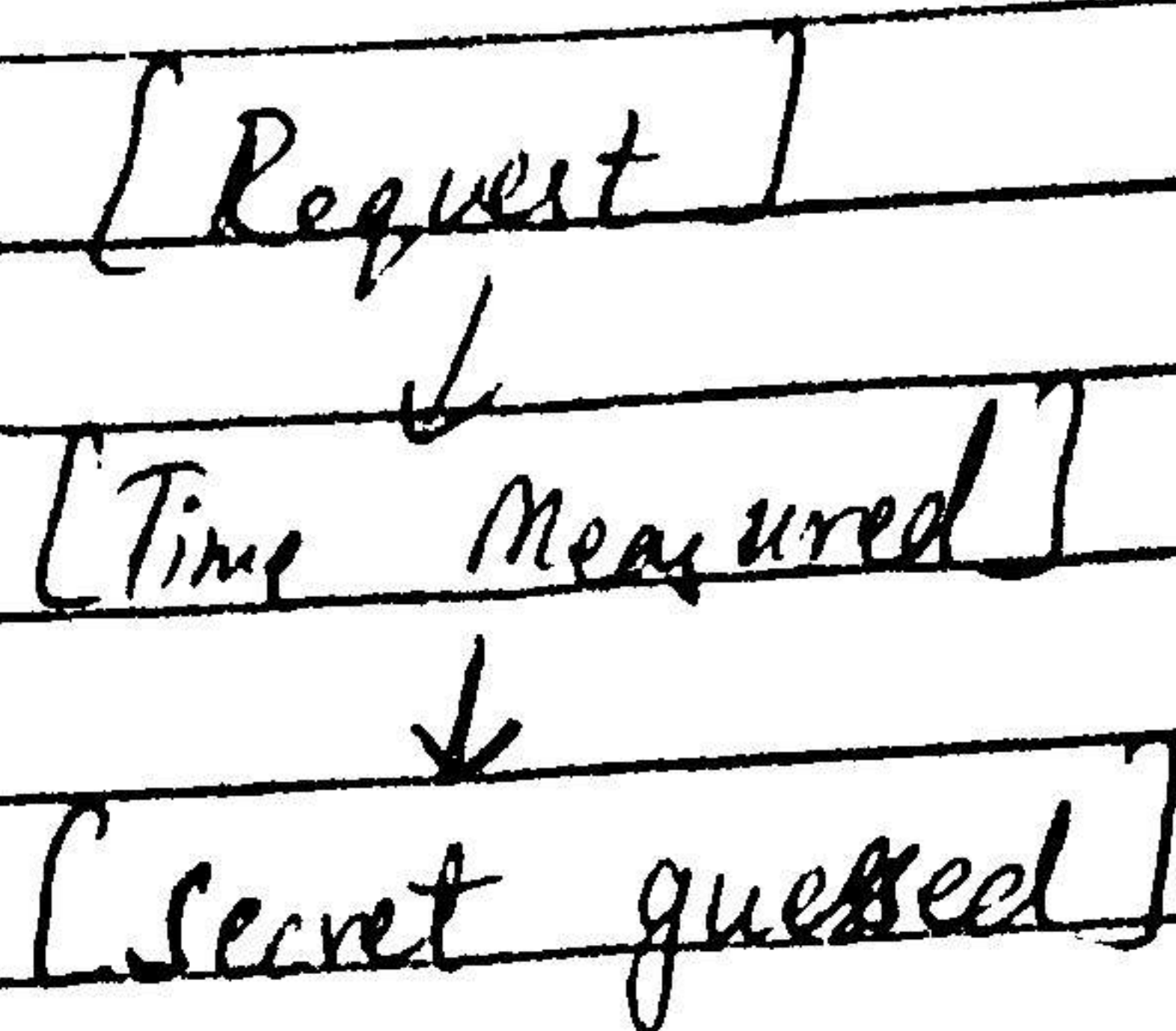


II) Timing Attack:-

Analyzing response time differences to extract sensitive data.

Security: Use constant-time algorithms and secure cryptographic design.

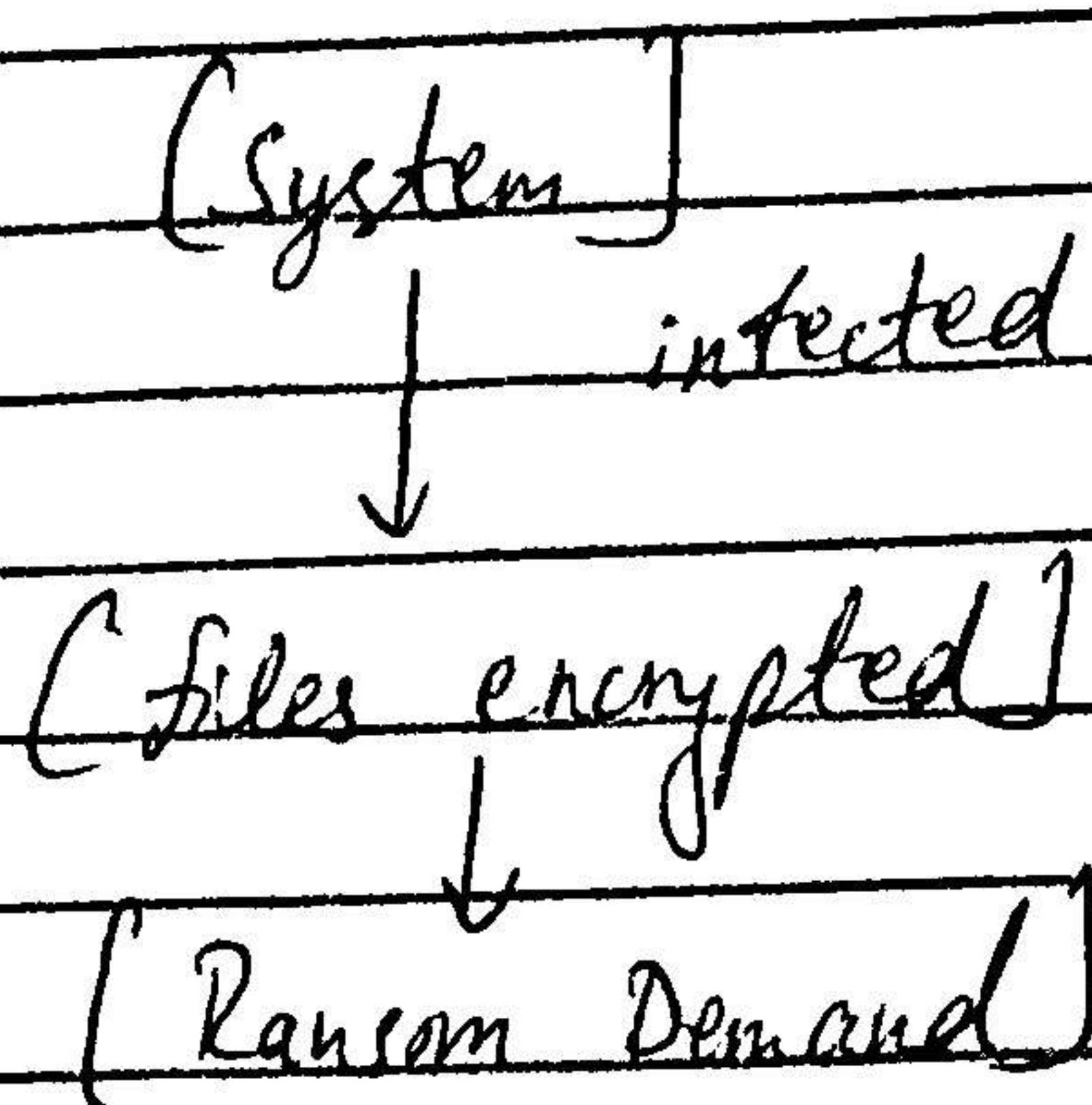
Environment: Cryptographic systems & APIs



12) Ransomware:-

Encrypt files and demands payment to regain access.

Security: Maintain backups and use endpoint protection
Environment: Enterprises and critical systems.

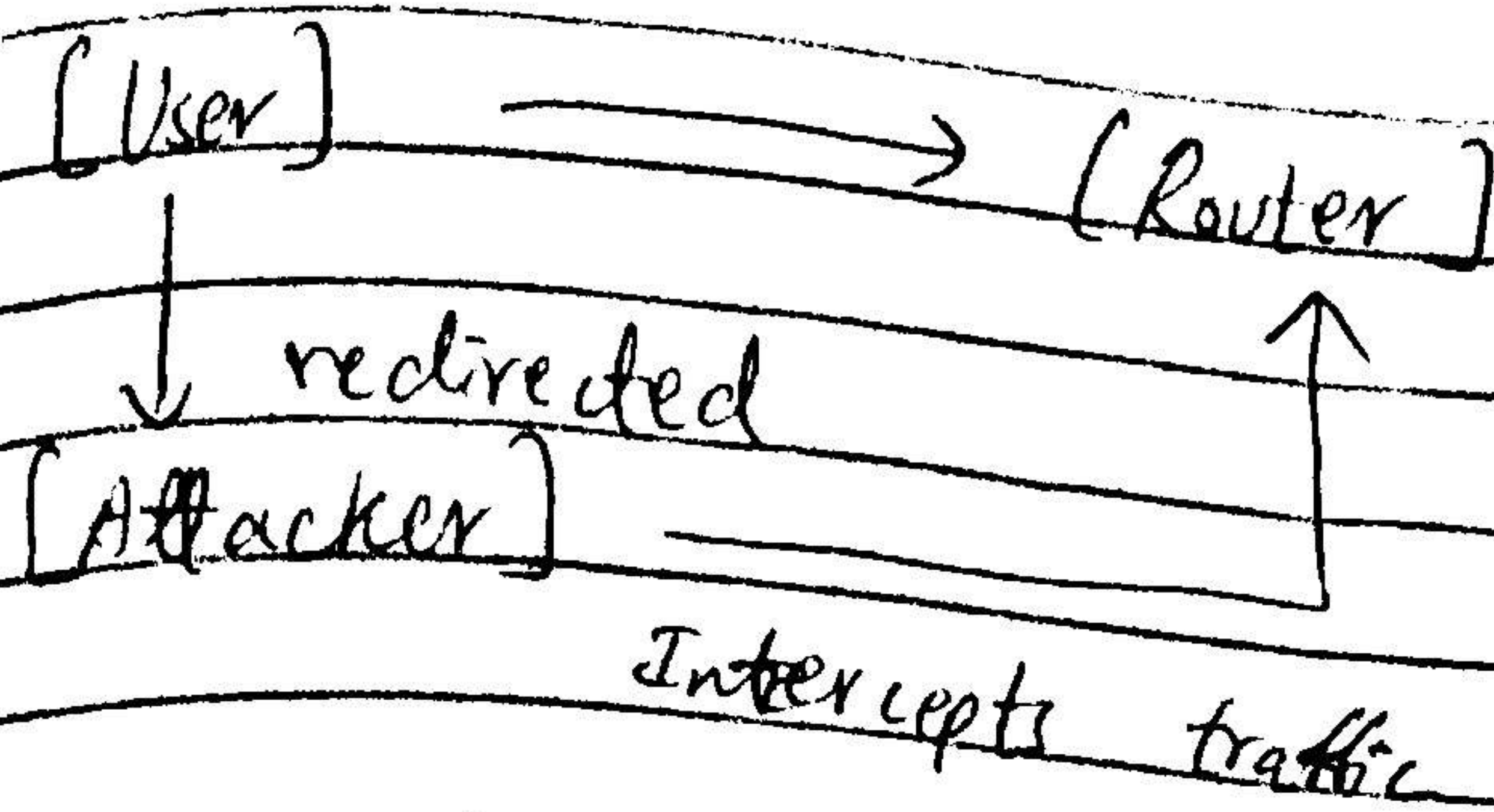


13) ARP Spoofing

The attacker sends fake ARP messages to redirect network traffic.

Security: secure switches and ARP monitoring tools

Environment: LAN

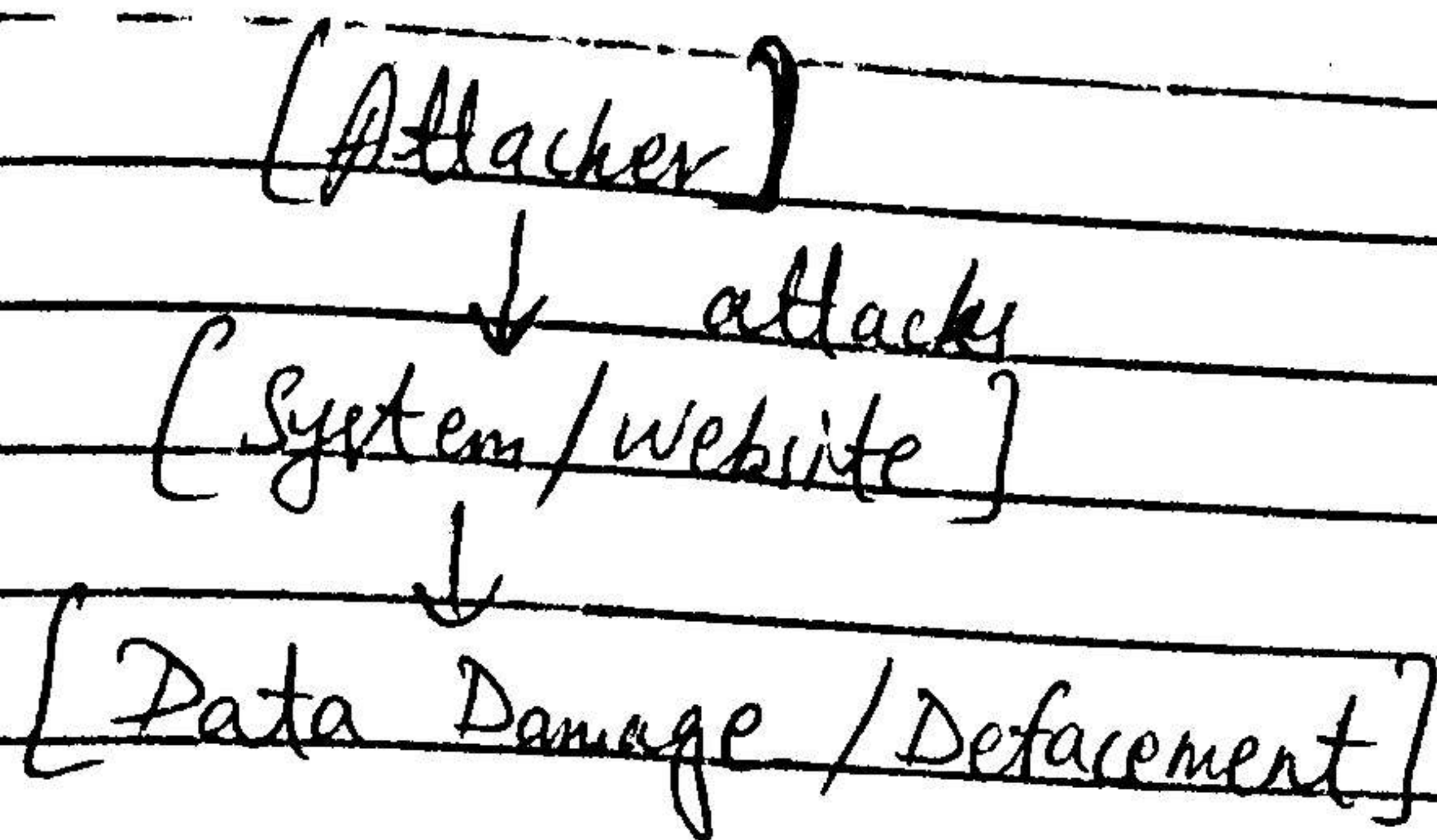


14) Sabotage/Vandalism

The attacker intentionally damages systems or defaces websites.

Security: Apply access control and monitoring

Environment: Web servers and internal systems.

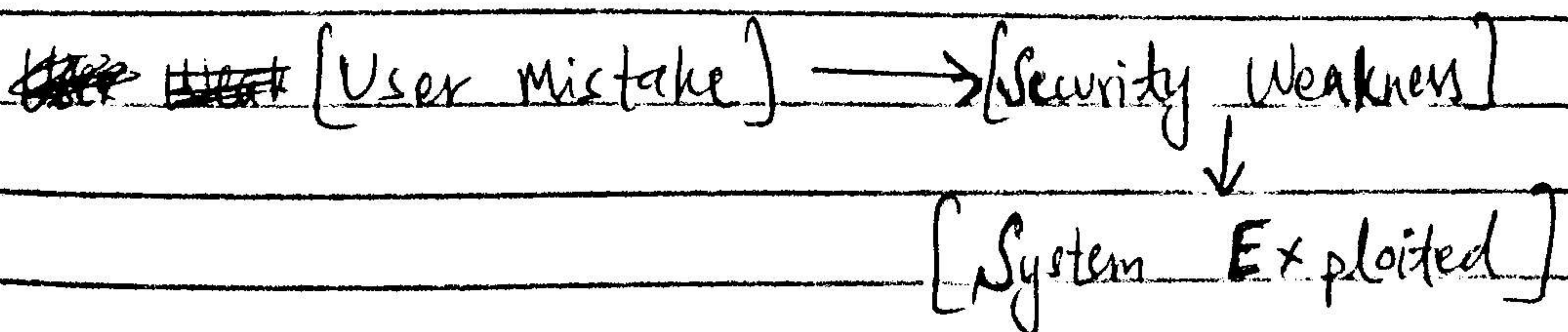


15) Human Error

Mistakes by users create vulnerabilities that attackers exploit.

Measures: Provide training and enforce strong policies

Environment: Any organization



16) Natural Attacks

Natural disasters damage infrastructure and disrupt services.

Security: Use disaster recovery plans and backups

Environment: Data centres and physical infrastructure.

[Natural Disaster]



[Infrastructure damage]



[System Downtime]